Digital Electronics Lab Report

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Experiment 2: Familiarization with 7493 (mod-16 counter) and 7490 (mod-10 counter).

Objectives:

1. To understand how to realize modulo-n (2<=n<=16) counter using 7493 or modulo-n (2<=n<=10) counter using 7490, and display counter states on a 7-segment LED
2. To construct a mod-60 or mod-100 counter using two 7493/7490 chips and display outputs on two 7-segment LED display blocks.

Apparatus:

1. 7493 or 7490 chips-2
2. 7-segment LED display blocks -Common Anode- 2

3. Resistor 470 Ω or 390 Ω -16
4. 7447 chips -2
5. Breadboard
6. DC power supply 5 V
7. Connecting wires

Theory:

Counters:

1. Counters are built using flip-flops.
2. Each flip-flop stores 1 bit.
3. If a counter has n flip-flops, it can represent 2^n states.
4. MOD-N counter - a counter that cycles through N states before repeating

Counters are sequential circuits that advance through a fixed number of states in response to clock pulses. The 74×93 is a 4-bit binary counter with a natural modulus of 16 because a 4 bit binary number can represent from 0 to 15. Smaller modulus counters are achieved by decoding specific states. Larger counters such as MOD-60 are realized by cascading a MOD-10 units counter with a MOD-6 tens counter where the carry output of the units stage provides the clock input to the tens stage. The 7490 is a BCD decade counter designed to count from 0 to 9. The resulting outputs are decoded using BCD-to-7-segment decoder ICs for human-readable display.

Pin Functions of 74xx93:

MR1 and MR2 are Reset Pins- Reset occurs only when both the pins are high.

CP0- is the primary clock input.

CP1- is the clock input for the 3 flip flops driven by Q0.

Q0,Q1,Q2,Q3 represent each bit where Q0 is LSB and Q3 is MSB. Vcc is connected to the 5V supply and gnd is grounded.

Pin Functions of IC 7447

BCD Inputs (A, B, C, D)- These inputs accept a 4-bit BCD number corresponding to decimal digits 0–9.

Segment Outputs (a–g)- These outputs drive the corresponding segments of the 7-segment display. A LOW output turns ON the segment.

Lamp Test (LT)- When LT = 0, all segments glow irrespective of the BCD input. It is used to check whether all LED segments are functioning properly.

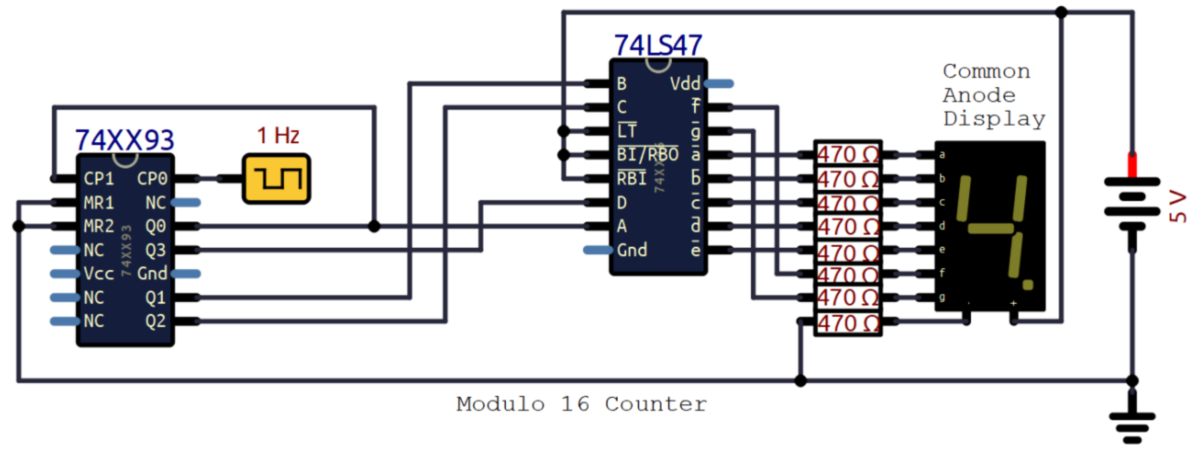
Ripple Blanking Input (RBI)- Used for leading zero suppression. When RBI is LOW and the BCD input is zero, the display is blanked.

Ripple Blanking Output (RBO)- Used in multi-digit displays. RBO of a less significant digit is connected to RBI of the next significant digit to suppress leading zeros.

Blanking Input (BI/RBO)- When BI is LOW, the entire display is turned OFF.

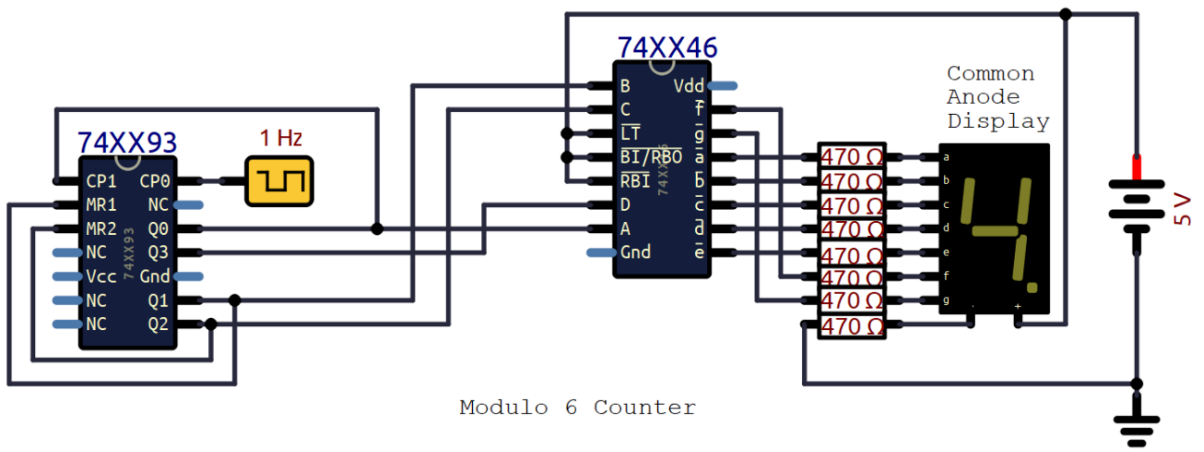
Circuit Diagram:

Mod 16 Counter:



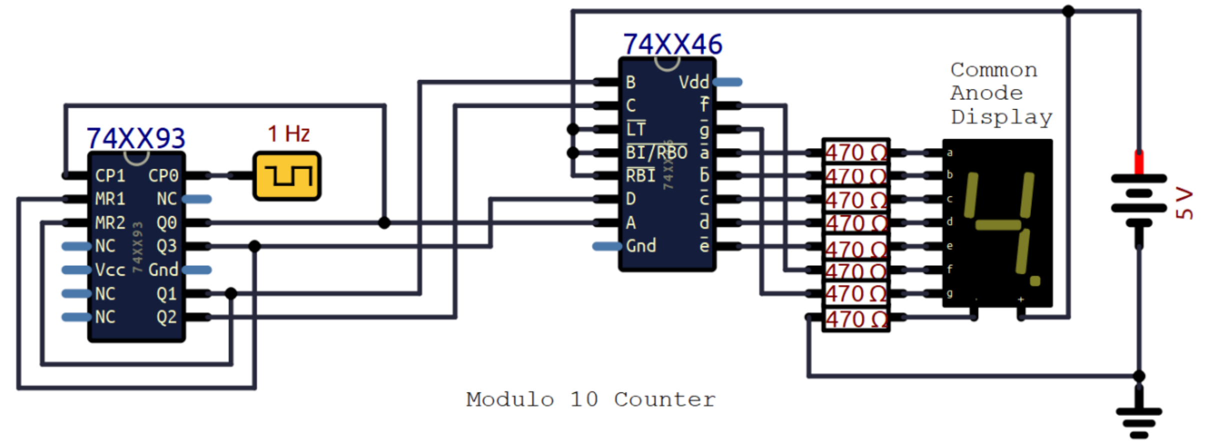
Counts from 0 to 15 and resets after 15

Mod 6 Counter:



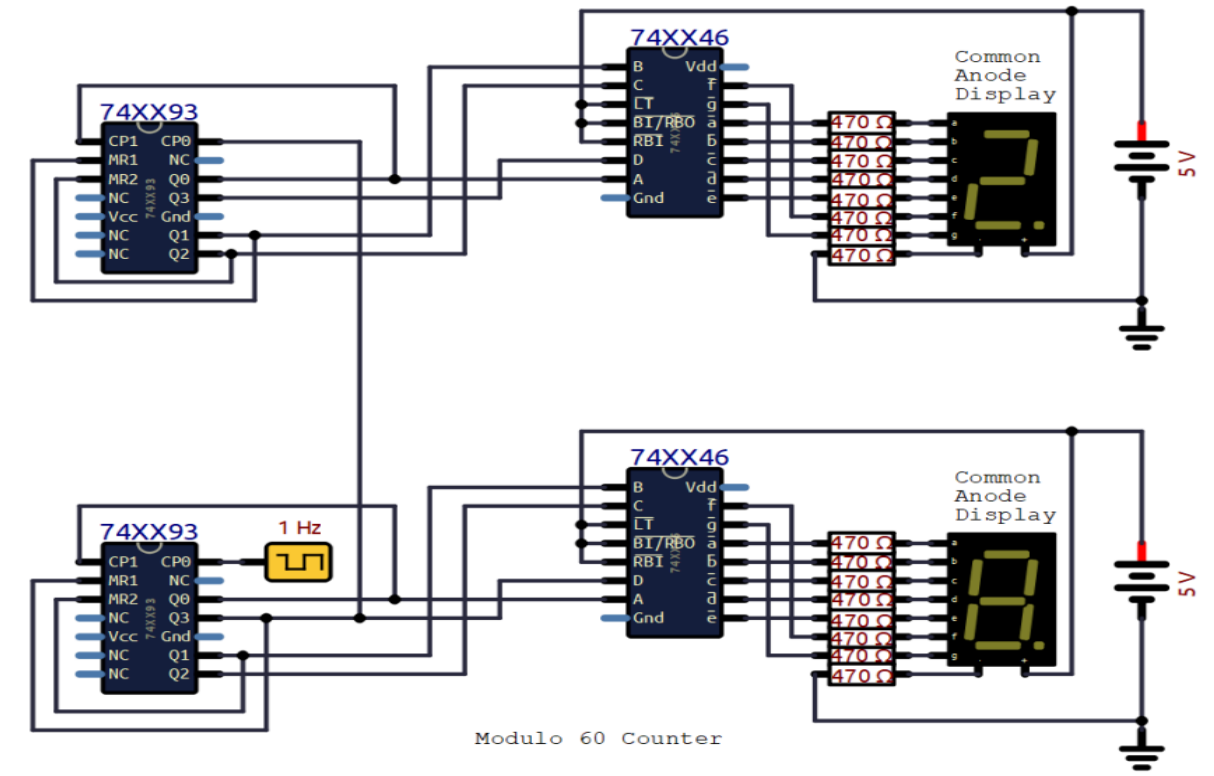
Counts from 0 to 5 and resets after 5. Binary code for 6 is 0110 so reset pins are connected to Q1 and Q2.

Mod 10 Counter:



Counts from 0 to 9 and resets after 9. Binary code for 10 is 1010 so reset pins are connected to Q1 and Q3.

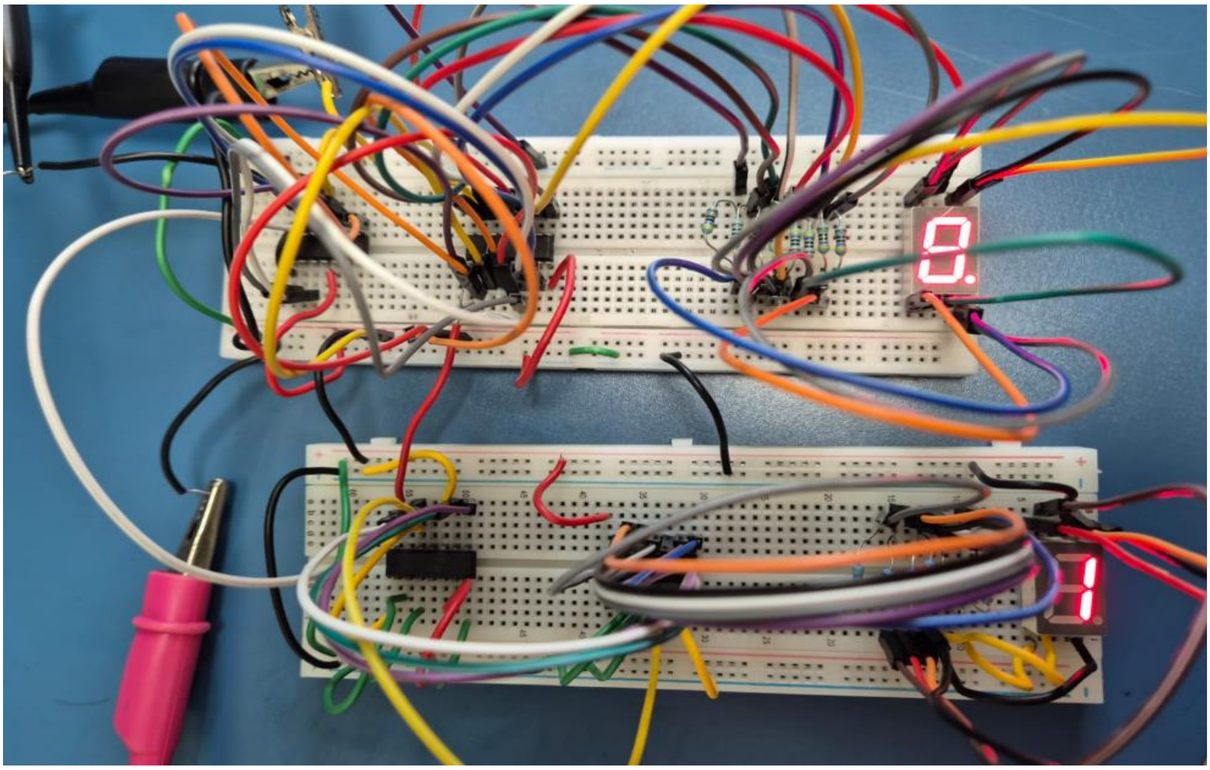
Mod 60 Counter:



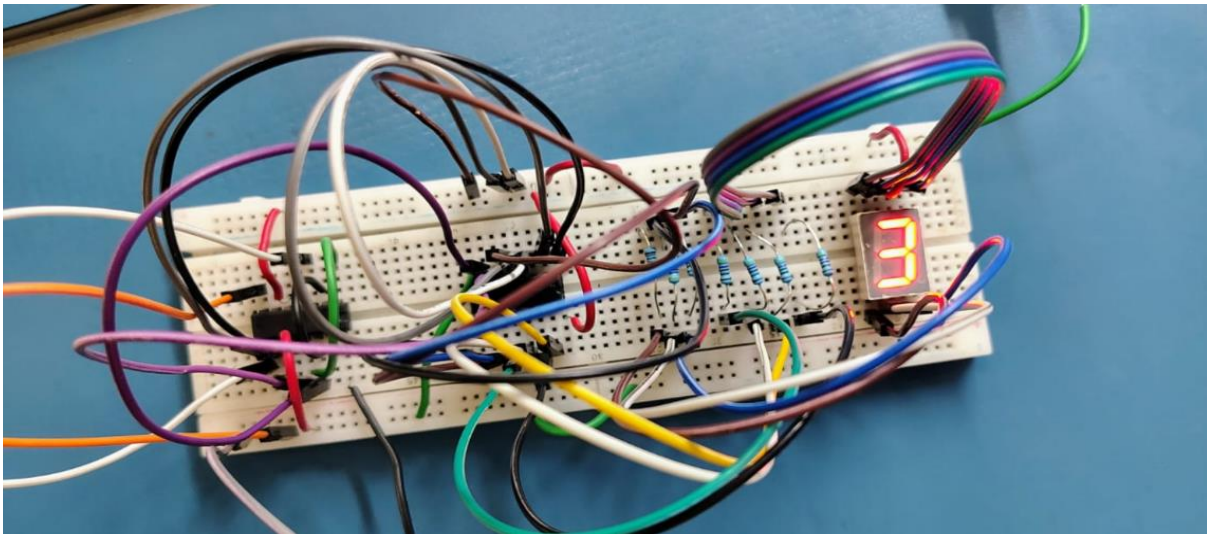
Done by cascading mod 6 and mod 10.

Observation:

Mod 60 Counter:



Mod 5 Counter:



Conclusion:

MOD-6, MOD-10, MOD-60, and MOD-5 were successfully implemented using counters by appropriate reset decoding and cascading techniques.

It was observed that:

1. The natural modulus of a counter is determined by the number of flip-flops (e.g., 4 flip-flops → MOD-16).
2. Lower moduli are obtained by detecting specific binary states and applying asynchronous reset.
3. Higher moduli such as MOD-60 are realized by cascading smaller counters and using carry propagation.
4. Changing only the reset pin connections allows the same counter hardware to be reconfigured for different counting ranges (e.g., MOD-6 to MOD-5).

This experiment demonstrates the flexibility of digital counters, the importance of pin-level understanding, and forms the foundation for practical systems such as timers, clocks and finite state machines.